



THE UNIVERSITY OF
MELBOURNE

SWEN90016

Software Processes and Management

Semester 1, 2026 Lecture 1

*Faculty of Engineering and Information Technology
School of Computing and Information Systems*

Week 1



Projects

1. Explore the key drivers of why projects fail / succeed.
2. Understand how organisations select the best / right projects (Project Screening).
3. Understand the Project Initialization process, Business Case structure and why organisations use them.
4. Explore various investment techniques and financial models.
5. Understand what a Project Charter is and how it is used.



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Example Project 1: Success or Failure – You decide



Failure or Success?

- Original estimate
 - \$500 million
- Final outcomes
 - \$1.35 billion
 - more than 5 years



Example 1: Failure ... Success

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Example Project 1: Success or Failure – You decide

Failure or Success?

- Original estimate
 - \$500 million
- Final outcomes
 - \$1.35 billion
 - more than 5 years



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Example Project 2: Success or Failure – You decide

Failure or Success?

- Original estimate
 - \$7m
 - 6 years
- Final outcomes
 - \$102m (1,357% more)
 - 16 years (+10 years)



Example 2: Failure ... Success

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Week 1

Example Project 2: Success or Failure – You decide



Failure or Success?

- Original estimate
 - \$7m
 - 6 years
- Final outcomes
 - \$102m (1,357% more)
 - 16 years (+10 years)



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Software Projects

History tells us we have failed.

ALL IT PROJECTS					
	2011	2012	2013	2014	2015
Successful	29%	27%	31%	28%	29%
Challenged	49%	56%	50%	55%	52%
Failed	22%	17%	19%	17%	19%

- **Successful:** project is completed on-time and on-budget, with all features and functions as initially specified.
- **Challenged:** completed and operational but over-budget, over the time estimate or offers fewer features and functions than planned.
- **Failed:** project is cancelled at some point during the development cycle.



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Software Projects - What determines success?

Success Factors	%
1. Executive Sponsorship	15%
2. Emotional Maturity	15%
3. User Involvement	15%
4. Optimisation – Statement of Requirements	15%
5. Skilled Resources	10%
6. Standard Architecture	8%
7. Agile Process	7%
8. Modest Execution	6%
9. Project Management Expertise	5%
10. Clear Business Objectives	4%

- Factors have remained relatively constant
- If we know the reasons, why can't we fix / improve it?
- 60% (first 4) are nontechnical items and difficult to change
- Broader organisational context and system at play

Standish Group 2015 CHAOS Report QA [Readings Online in CANVAS]

The Curious Case of the CHAOS Report 2009 [Readings Online in CANVAS]



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Digital transformation projects - some stories about why projects failed

GE created a new digital business unit but was focused on size instead of quality

Ford started a new digital service that was separate from the rest of the company instead of integrating digital solutions

Procter & Gamble didn't consider the competition or impending economic crash

[Companies That Failed At Digital Transformation And What We Can Learn From Them \[Readings Online in CANVAS\]](#)



Week 1



Week 1 – Intended Learning Objectives

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Project Screening and where to start

“If you don’t know where you’re going, any road will take you there”.

Any Road by George Harrison – The Beatles

- The place to start is at the beginning!
- Organisations need a formal, structured approach to:
 - Select;
 - Prioritise;
 - Have oversight; and
 - Drive accountability across all projects.

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Setting up a project for success. A Business Case is the key.

The purpose of the Business Case is to establish mechanisms to judge whether the project is (and remains) desirable, viable and achievable to support decision making in its initial and continued investment.

- Provides a factual base for key decisions makers to decide if the project should be undertaken
- Demonstrates how the project adds value to the organisation
- Has a set of pre-defined standard organisational characteristics (costs, benefits, risk, etc.)
- It is not all about size - size depends on the cost / benefit
- It is a living document throughout the project that should be reviewed and signed off at key stages



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Setting up a project for success. The Business case is key

Business case contains:

- Executive summary
- Reasons / explanation of why it is required
- Business options
- Expected benefits
- Expected dis-benefits
- Timescale
- Costs
- Investment appraisal
- Major risks

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Business Case. Who is responsible for what?

Role	Responsibilities
Corporate	1. Provides Mandate / The go ahead. 2. Holds Senior Users accountable for benefits realisation. 3. Responsible for conducting post projects benefits validation.
Executive / Sponsor	1. Owns the Business Case. 2. Responsible for reviewing the benefits throughout the project.
Senior Users	1. Responsible for accepting the benefits and delivering them. 2. Responsible for ensuring the delivered products are to the appropriate quality standard. 3. Provides on-going actual V forecasted benefit realisation.
Project Manager	1. Prepares the Business Case. 2. Conducts Risk assessment and impact analysis. 3. Assess and updates the Business Case at each defined stage.
Project Assurance / QA	1. Assists in developing the Business Case. 2. Ensure value for money and risks are continuously managed. 3. Monitors change to the Business Case and validates it.
Project Support	1. Responsible for capturing data and preparing management reports. 2. Key support point for all project stakeholders – schedules, cost analysis, minutes, actions, supplier liaison etc.

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Nobody has responded yet.

Hang tight! Responses are coming in.

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It is all about the money!

- For non mandatory projects, the primary benefit is (*often*) financial
- Multiple investment techniques are used to analyse the investment required / financial benefit
- Some (there are many more) techniques include:
 - *Return On Investment*
 - Net Present Value
 - Payback period
 - Rough Order of Magnitude

However, it is not always about the best return – organisations need to invest in all parts of their business



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Investment Techniques – Return On Investment (ROI)

- ROI is income divided by investment
 - $ROI = (\text{total discounted benefits} - \text{total discounted costs}) / \text{total discounted costs}$
- “discounted” here means adjusted to reflect value in current terms
- The higher the ROI % or higher the ratio of benefits to costs, the better it is
- Many organisations have a required rate of return or minimum acceptable rate of return on investment for projects (11% to 14%)

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It all begins with a Project Charter



[Project planning in Project Management \[Readings Online in CANVAS\]](#)



Week 1 – A look at the Project Charter



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